PAPER ID-311195

Roll No: | | | | | | | | | | | | | |

**BTECH**
**(SEM V) THEORY EXAMINATION 2025-26**
**INTRODUCTION TO DATA ANALYTICS AND VISUALIZATION**

**TIME: 3 HRS** **M.MARKS: 70**

**Note:** Attempt all Sections. In case of any missing data; choose suitably.

## SECTION A

**1. Attempt *all* questions in brief.** **02 x 7 = 14**

| Q no. | Question | CO | Level |
|---|---|---|---|
| a. | Define Data Analytics and explain its importance in decision making. | 1 | K2 |
| b. | Differentiate between structured, semi-structured and unstructured data. | 1 | K2 |
| c. | List and briefly explain the phases of Data Analytics lifecycle. | 2 | K2 |
| d. | What is regression modeling? Mention one application. | 2 | K3 |
| e. | Define data streams with one real-time example. | 3 | K2 |
| f. | What is frequent itemset mining? | 4 | K3 |
| g. | State two principles of human vision used in data visualization. | 5 | K2 |

## SECTION B

**2. Attempt any *three* of the following:** **07 x 3 = 21**

| | | | |
|---|---|---|---|
| a. | Explain sources and characteristics of data with suitable examples. | 1 | K2 |
| b. | Explain regression modeling and discuss its limitations. | 2 | K2 |
| c. | Describe stream data model and architecture with neat diagram. | 3 | K3 |
| d. | Explain Apriori algorithm and its role in frequent itemset mining. | 4 | K3 |
| e. | Explain challenges involved in effective data visualization. | 5 | K3 |

## SECTION C

**3. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Explain Data Analytics lifecycle with a neat diagram and example. | 1 | K2 |
| b. | Analyze the role of data analytics in modern organizations. | 1 | K3 |

**4. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Explain Bayesian modeling and in data analytics. | 2 | K2 |
| b. | Compare regression modeling and Bayesian modeling. | 2 | K3 |

**5. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Explain sampling and filtering techniques in data streams. | 3 | K2 |
| b. | Analyze challenges in mining high-speed data streams. | 3 | K3 |

**6. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Explain clustering techniques used in data mining. | 4 | K2 |
| b. | Analyze the working of K-means clustering with an example. | 4 | K3 |

**7. Attempt any *one* part of the following:** **07 x 1 = 07**

| | | | |
|---|---|---|---|
| a. | Explain stages of data visualization. | 5 | K2 |
| b. | Evaluate the impact of human vision limitations on visualization design. | 5 | K3 |

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